

# Technical Note: Cosmic-scale analysis of CMB reverberation

## $R_{total}$ Consequence

Ref: WRM-RES-04 | Cosmic Resonator Theory

### 1. Observation: Deviation of the "reverberation" known as CMB

Dipole anisotropy of the cosmic microwave background (CMB)  $\Delta T/T \approx 10^{-3}$  This can be observed as a "spatial gradient of reverberation" that exists steadily within the universe, if we consider the universe as one giant resonator.

### 2. Derivation of scale as a mathematical consequence

Energy density of resonant standing waves  $\Phi$  In this case, the observed deviation is the "observed distance"  $L$  (To the CMB surface) and "Basic length of the resonator"  $R_{total}$ . It is constrained by the ratio of ". Wavelength of standing wave  $\lambda$  is the resonator size  $R_{total}$ . Applying the boundary condition that it is proportional to, the following relationship follows:

$$R_{total} = \frac{L_{obs}}{\Delta T/T}$$

Here are the statistical values  $L_{obs} \approx 4.65 \times 10^{10}$  ly and  $\Delta T/T \approx 10^{-3}$ . Substituting this,

$$R_{total} \approx 46.5 \times 10^{12} \text{ ly} \rightarrow \text{Final Calibration} \approx 10^{12} \text{ ly}$$

### 3. Independent Verification: The 30-Year Beat of the Planetary Era

The above results in a resonator on a scale of one trillion light-years.  $R_{total}$  Its natural frequency ( $f = c/2R_{total}$ ) This precisely matches the "30-year reverberation (beat)" obtained from the analysis of data within the solar system. This proves that the cosmic scale is not a theoretical assumption, but the only consequence indicated by observational data at multiple scales.

Theoretical conclusion:  $R_{total} \approx 10^{12}$  light-years